

EXPLANATIONS

1. C_nH_{2n} is the general formula of alkenes. The second member of this series is propene having formula C_3H_6 .

(1 Mark)

2. The testes are the male reproductive organs in human beings and perform two vital functions:

- Produce male sex hormone (testosterone) (½ Mark)
- Produce male gamete (sperms) (½ Mark)

3. Ozone in the upper atmosphere absorbs and filters out harmful ultraviolet (UV) radiation from the sun, protecting life on earth from its harmful effects such as skin cancer.

(1 Mark)

4. The four characteristics of the images formed by plane mirrors are:

- (i) The size of the images is identical to that of the objects. (½ Mark)
- (ii) The images are produced behind the mirrors at the equivalent distance from the mirrors as the objects. (½ Mark)
- (iii) The images formed are erect and virtual. (½ Mark)
- (iv) The images undergo lateral inversion. (½ Mark)

5. *Out of the Syllabus*

6. *Out of the Syllabus*

7. The process of adding hydrogen to unsaturated compounds is known as hydrogenation. (½ Mark)

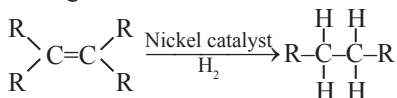
Example: $CH_2 = CH_2 + H_2 \rightarrow C_2H_6$ (½ Mark)

Essential Conditions:

Presence of unsaturated compound

Presence of Ni or Pd catalyst (1 Mark)

When vegetable oils (containing long unsaturated carbon chains) undergo hydrogenation, formation of animal fats having saturated carbon chains takes place.



(1 Mark)

8. Soaps molecules are generally sodium or potassium salts of long chain carboxylic acids. (½ Mark)

Detergent molecules are generally sodium salts of ammonium or sulphonate salts with chloride or bromide ions. (½ Mark)

Cleansing action of soap:

Most of the dirt is oily in nature and oil does not dissolve in water. An ionic end of soap interacts with water while carbon chain interacts with oil. At this stage, formation of

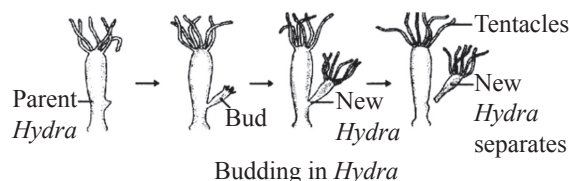
micelles takes place in which one end of the molecules is towards the oil while the other end faces outside. This results in formation of emulsion in water and hence, soap micelle helps to pull out the dirt in water. (2 Marks)

9. *Out of the Syllabus*

10. *Out of the Syllabus*

11. DNA copying is a process that occurs before cell division, where a cell makes a similar copy of its DNA. The mechanism involves the unwinding and separation of the double-stranded DNA molecule, synthesis of new complementary strands, and the addition of nucleotides based on the base-pairing rules. DNA copying mechanisms is not absolutely reliable and hence it is only to be expected that the process of copying of the DNA will have some variations each time. After cell division, two resultant cells are similar but subtly different from each other. This inbuilt tendency for variation during reproduction is the basis for evolution. (3 Marks)

12. Budding is a type of asexual reproduction in *Hydra*, where a small bud or outgrowth forms on the parent *Hydra* body, which grows into a fully formed daughter *Hydra*. The daughter *Hydra* develops tentacles, a mouth, and other body structures, and eventually separates from the parent *Hydra* to become an independent organism capable of feeding and reproducing on its own.



(3 Marks)

13. Four methods of contraception commonly used by humans are:

Barrier methods: Such as condoms, diaphragms, and cervical caps, which physically prevent the sperm from reaching the egg.

Hormonal methods: Such as birth control pills, patches, which contain synthetic hormones that prevent ovulation or thicken the cervical mucus.

Intrauterine devices (IUDs): Small, T-shaped devices that are inserted into the uterus to prevent fertilisation or implantation.

Sterilization: Such as vasectomy in males and tubectomy in females, which permanently prevent pregnancy by cutting and blocking the tubes that carry the sperm or eggs. (2 Marks)

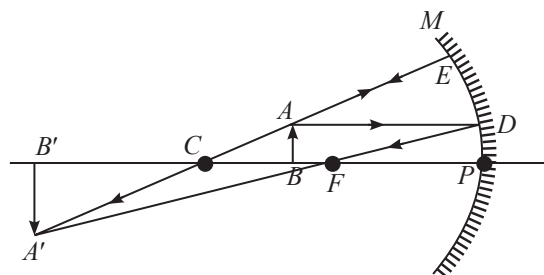
Use of contraception can have a direct effect on the health and prosperity of a family:

- Family planning and spacing pregnancies
- Reduce maternal/infant mortality risk (1 Mark)

14. Out of the Syllabus

15. Out of the Syllabus

16. The two chosen rays are given in the ray diagram below.



(½ Mark)

- (i) A parallel ray to the principal axis, upon reflection, will go through the principal focus of a concave mirror.

(½ Mark)

- (ii) A ray that passes through the centre of curvature of a concave mirror will be reflected back along its original path after reflection.

(½ Mark)

Now, use the mirror formula to find out the image distance:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad (\frac{1}{2} \text{ Mark})$$

Now, substitute the provided values in the above formula:

$$\frac{1}{-10} = \frac{1}{-15} + \frac{1}{v} \Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{10}$$

$$\frac{1}{v} = \frac{-3+2}{30} = -\frac{1}{30} \Rightarrow v = -30 \text{ cm} \quad (\frac{1}{2} \text{ Mark})$$

The formation of the image occurs 30 cm in front of the mirror (the negative sign indicating that the object and image are on the same side). The image is real and inverted.

(½ Mark)

17. Out of the Syllabus

18. (a) Burning plastic releases harmful gases and toxic chemicals into the atmosphere, which can cause air pollution and harm the environment. It also contributes to the green house effect and climate change. Therefore, burning plastic is not an eco-friendly method of waste disposal. (1 Mark)

Rakesh's method of separating biodegradable waste (left over food and fruit peels) from non-biodegradable waste (plastic) was the excellent choice because this practice will help easy disposal and treatment of waste. Plastic waste would go for recycling while left over food and fruit peel waste would go for the creation of a compost. (1 Mark)

- (b) To contribute to maintaining the parks and roads neat and clean, we can follow these steps:

- Dispose off waste in designated bins.
- Avoid littering.

- Participate in clean-up drives organised by the local authorities. (1 Mark)

19. Carbon contains four valence electrons. It is difficult to lose or gain 4 electrons to achieve noble gas configuration. Hence, it will share its four valence electrons with other atoms of carbon or with atoms of other elements. Hence, carbon forms compounds mainly by covalent bonds.

(1½ Marks)

- (i) Carbon has the unique ability to form bonds with other atoms of carbon which gives rise to large molecules. This property is known as catenation. No other element exhibits the property of catenation to the extent seen in carbon. (1 Mark)

- (ii) Carbon has 4 valence electrons due to which it can form bonds with four other C atoms or atoms of other mono-valent elements. (1 Mark)

Bonds formed by carbon with other elements are very strong which makes those compounds exceptionally stable. The reason for this is due to the small size of carbon due to which the nucleus can hold a shared pair of electrons strongly. (1½ Marks)

20. Ovary: The ovary is a pair of reproductive glands that produce and release eggs through a process called ovulation. The ovaries also produce hormones estrogen and progesterone, which play a crucial role in regulating the menstrual cycle and the development of female sex characteristics.

Oviduct: The oviduct, also known as the fallopian tube, is a pair of tubes that connect the ovary to the uterus. The oviduct is a site where fertilization occurs, as the sperm meet the released egg and fertilise it. The oviduct also helps in transporting the fertilized egg to the uterus.

Uterus: The uterus is a muscular organ that provides a nourishing environment for the developing embryo. During pregnancy, the fertilised egg implants in the uterus, and the uterus provides a nurturing environment for the developing foetus. The uterus also contracts during childbirth to help expel the baby through the birth canal.

(1 × 3 Marks)

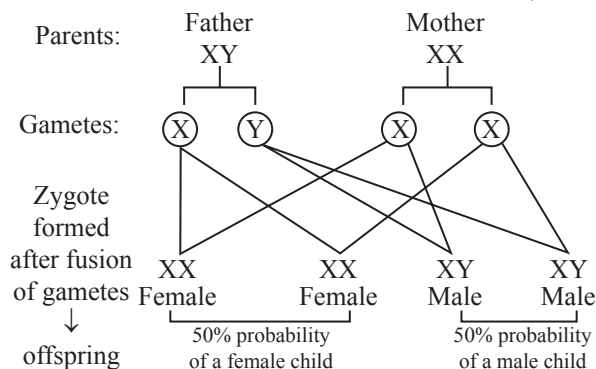
The embryo receives nourishment inside the mother's body through the placenta, which develops in the uterus during pregnancy. Placenta contains villi on the embryo's side of the tissue. On the mother's side are blood spaces, which surround the villi. This provides a large surface area for glucose and oxygen to pass from the mother to the embryo. The developing embryo will also generate waste substances which can be removed by transferring them into the mother's blood through placenta. (2 Marks)

21. Human beings typically have 23 pairs of chromosomes.

(1 Mark)

Out of the 23 pairs of chromosomes in a human being, one pair is the sex chromosomes. (1 Mark)

Females have two X chromosomes (XX), and males have one X and one Y chromosome (XY). The presence or absence of the Y chromosome determines the biological sex of an individual. (1 Mark)



As depicted in the flow chart, the sex of a newborn child is determined by the random fusion of sperm and egg during fertilization, which results in the formation of an XX or XY zygote. During fertilization, the sperm carries either an X or a Y chromosome, determining the sex of the offspring. If the sperm carries an X chromosome, the resulting zygote will be XX (female). If the sperm carries a Y chromosome, the resulting zygote will be XY (male). (1 Mark)

22. (a) The two laws of refraction of light are:

- The incident ray, refracted ray, and the normal at the point of incidence are all coplanar. (½ Mark)
- At a point in a medium the ratio of sine of angle of incidence to the sine of angle of refraction is constant. (½ Mark)

The ratio of the speed of light in a vacuum to the speed of light in a provided medium is termed as the absolute refractive index.

$$\text{Refractive index} = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

$$n = \frac{c}{v} \quad (1 \text{ Mark})$$

(b) (i) Now, to calculate the speed of light in vacuum use the formula of refractive index:

$$n_B = \frac{c}{v_B} \Rightarrow c = n_B \times v_B$$

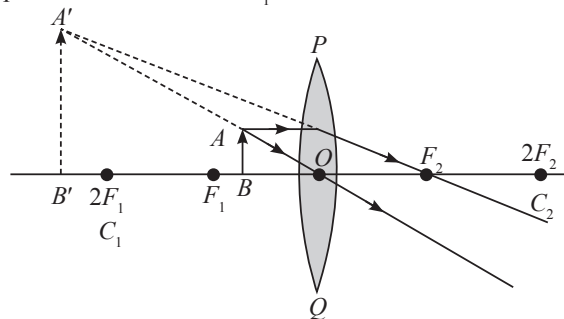
Now, substitute the values in the above equation:
 $c = 1.5 \times 2 \times 10^8 = 3 \times 10^8 \text{ m/s}$ (1½ Marks)

(ii) Use the same formula to calculate the speed of light in medium 'A':

$$n_A = \frac{c}{v_A} \Rightarrow v_A = \frac{c}{n_A}$$

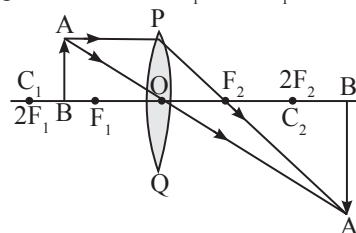
$$v_A = \frac{3 \times 10^8}{2} = 1.5 \times 10^8 \text{ m/s} \quad (1\frac{1}{2} \text{ Marks})$$

23. Magnified erect image of the object, here the object is placed between O and F_1



(1½ Marks)

Magnified inverted image of the object, here the object is placed between F_1 and $2F_1$



(1½ Marks)

Apply the lens formula:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

Substitute the given value with sign conventions in the above equation:

$$\frac{1}{v} = -\frac{1}{10} - \frac{1}{20} = -\frac{3}{20} \Rightarrow v = -\frac{20}{3} = -6.6 \text{ cm} \quad (2 \text{ Marks})$$

24. Myopia, also known as nearsightedness, is a condition where the human eye can see nearby objects clearly but struggles to focus on distant objects. (1 Mark)

Causes:

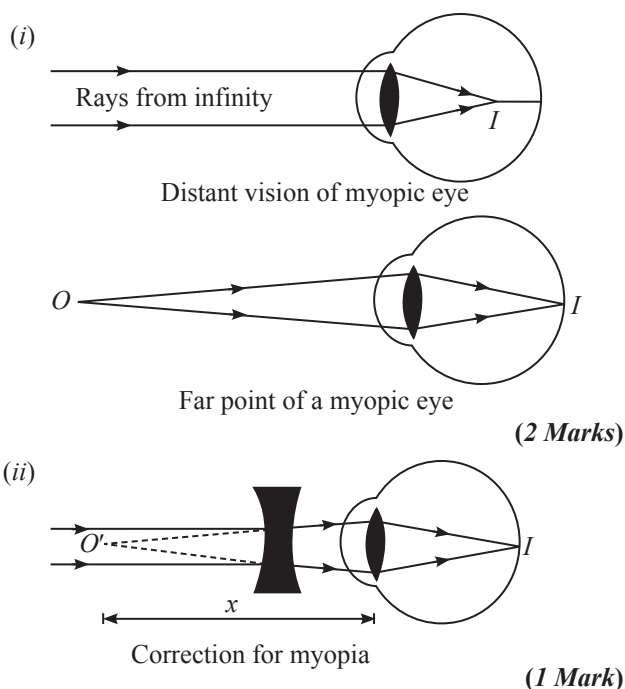
Increase in eyeball length: The distance between the retina and the eye lens becomes greater, leading to focusing issues with distant objects.

Decrease in eye lens focal length: The ciliary muscles holding the eye lens may not fully relax, causing tension and altering the lens shape. (½ Mark)

Correction:

This defect can be corrected by using a concave lens of suitable focal length. So, a man suffering from this defect wears spectacles having concave lens of suitable focal length. The concave lens diverges the rays of light entering the eye from infinity.

Focal length of concave lens used for correcting the myopic eye is equal to distance of far point of the myopic eye. (½ Mark)

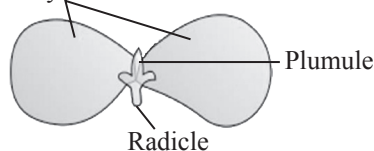


25. (a) The formation of clear transparent solution will take place as acetic acid gets completely dissolved in distilled water. (1 Mark)

26. (d) NaOH is a base and it gets readily dissolved in water which results in the formation of an alkaline solution. It will turn red litmus to blue. When NaOH is dissolved in water, a large amount of heat is evolved which means the beaker will appear hot when touched from outside. (1 Mark)

27. (c) Hard water contains chlorides, sulfates and bicarbonates of Ca and Mg. So, according to this, option (c) is correct. (1 Mark)

28. (b) Gram seed embryos contain cotyledon, plumule and radicle. (1 Mark)



29. (d) Out of the Syllabus

30. (c) Since the light rays originating from an infinite distance are reflected by object 'X' and meet at a point situated 12 cm away, it can be concluded that object 'X' is a concave mirror with a focal length of 12 cm. (1 Mark)

31. (b) Here, we are assuming the object distance as infinity. Now, apply the lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Since the object distance is at infinity. Therefore:

$$\frac{1}{u} = 0 \text{ and } f = v$$

The student should know the image distance to find the focal length of the lens. (1 Mark)

32. (c) The experiment conducted by student R demonstrates a systematic approach, and the relation behind it can be explained by applying Snell's law.

As per the Snell's Law:

$$\mu_{air} \cdot \sin(i) = \mu_g \cdot \sin(r)$$

$$(1) \cdot \sin(i) = (1.5) \cdot \sin(r)$$

$$\sin(r) = \frac{\sin(40)}{1.5} \Rightarrow (r) = \sin^{-1} 0.42 \Rightarrow r \approx 25^\circ \text{ (1 Mark)}$$

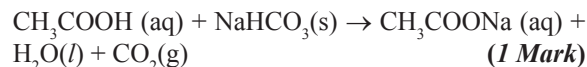
33. (b) $\angle i$ is the angle of incidence and $\angle e$ is the angle of emergence. (1 Mark)

34. Two observations are:

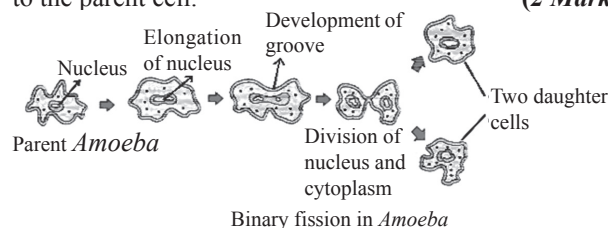
(i) Brisk effervescence (½ Mark)

(ii) Evolution of colourless and odorless gas (½ Mark)

The chemical reaction is:



35. The type of asexual reproduction that involves the formation of two individuals from a single parent and the loss of parental identity is called binary fission. This process is commonly seen in single-celled organisms, such as bacteria and protozoa. In binary fission, the parent cell replicates its genetic material and divides into two identical daughter cells. The initial stage of binary fission involves DNA replication and elongation of the cell. The final stage involves the separation of the two daughter cells, which become independent and genetically identical to the parent cell. (2 Marks)



36. (a) When the student adjusts the position of the object closer to the lens, the image moves farther away from the lens. To achieve a clear image, the student should adjust the position of the screen away from the lens. (½ Mark)

(b) Moving the object closer to the lens results in an enlargement of the image. (½ Mark)

(c) If the object is positioned in close proximity to the lens, it can be considered to be located between the focal point and the optical centre. In such a scenario, the resulting image is virtual, erect, and enlarged. (1 Mark)